Band-Limitedness of the Warped Pullback of the Hardy Z-Function

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Abstract

Let Z denote the Hardy Z-function and $\Theta(t) = \vartheta(t) + ct$ with $c > c_\star := -\inf_{\mathbb{R}} \vartheta'$. The warped pullback

$$Y(u) := Z(\Theta^{-1}(u))/\sqrt{\Theta'(\Theta^{-1}(u))}$$

has windowed Fourier transform $\widehat{Y_T}$ satisfying $|K_T(\mu)|^2/\Theta(T) = O_\mu(T^{-1/2})$ for every fixed $|\mu| > 1$, where $K_T = \widehat{Y_T}/(2\pi)$ and $\Theta(T) = \frac{1}{2}T\log(T/2\pi) + O(T)$. The proof uses Gabcke's second-order Riemann–Siegel remainder bound $R_{\mathrm{RS}}(t) = O(t^{-3/4})$. Consequently the windowed second-moment spectral measures $d\rho_T(\mu) = |\widehat{Y_T}(\mu)|^2/(2\pi\Theta(T))\,d\mu$ concentrate on $[-1, 1]$ with total mass tending to 2, and any weak-$*$ subsequential limit ρ has $\operatorname{supp}\rho \subseteq [-1, 1]$.

1 Setup

Definition 1 (Hardy function)**.** $\vartheta(t) := \arg\Gamma(\frac{1}{4} + \frac{it}{2}) - \frac{t}{2}\log\pi$ (continuous branch, $\vartheta(0) = 0$), and $Z(t) := e^{i\vartheta(t)}\zeta(\frac{1}{2} + it)$, real-valued on $\mathbb{R}$.

Definition 2 (Warp)**.** Fix $c > c_\star := -\inf_{t\in\mathbb{R}} \vartheta'(t)$. Set $\Theta(t) := \vartheta(t) + ct$. Then $\Theta'(t) = \vartheta'(t) + c \geq c - c_\star > 0$ on $\mathbb{R}$, so $\Theta : \mathbb{R} \to \mathbb{R}$ is a C^∞ diffeomorphism.

Definition 3 (Warped pullback)**.** $Y(u) := Z(\Theta^{-1}(u))/\sqrt{\Theta'(\Theta^{-1}(u))}$ for $u \in \mathbb{R}$, principal branch of the positive square root.

Definition 4 (Windowed Fourier transform)**.** For $T \geq 2$ and $\mu \in \mathbb{R}$,

$$\widehat{Y_T}(\mu) := \int_0^{\Theta(T)} Y(u)\, e^{-i\mu u}\, du, \qquad K_T(\mu) := \widehat{Y_T}(\mu)/(2\pi).$$

Axioms used. **(RS)** Riemann–Siegel with Gabcke remainder [2]: $Z(t) = 2\sum_{n\leq N(t)} n^{-1/2}\cos(\vartheta(t) - t\log n) + R_{\mathrm{RS}}(t)$ with $N(t) = \lfloor\sqrt{t/2\pi}\rfloor$ and

$$R_{\mathrm{RS}}(t) = O(t^{-3/4}),$$

obtained by retaining the second-order term of the Riemann–Siegel asymptotic series. Gabcke's 1979 thesis establishes the bound with an explicit effective constant.

(HL) Hardy–Littlewood: $\int_0^T |Z|^2\, dt = T\log(T/2\pi) + (2\gamma - 1)T + O(T^{1/2}\log T)$.

(ϑ) $\vartheta'(t) = \frac{1}{2}\log(t/2\pi) + O(t^{-2})$; $\vartheta''(t) = 1/(2t) + O(t^{-3})$.

2 Plancherel mass

Lemma 5 (Plancherel via warp). $\int_0^{\Theta(T)} |Y(u)|^2\,du = \int_0^T |Z(t)|^2\,dt$*, hence*

$$\int_{\mathbb{R}} |\widehat{Y_T}(\mu)|^2\,d\mu = 2\pi \int_0^T |Z|^2\,dt.$$

Proof. Change of variables $u = \Theta(t)$, $du = \Theta'(t)\,dt$: $\int |Y|^2 du = \int (|Z|^2/\Theta')\Theta'\,dt = \int |Z|^2\,dt$. The second identity is Plancherel on $L^2([0,\Theta(T)])$. □

Corollary 6 (Total mass). $\frac{1}{2\pi\Theta(T)} \int_{\mathbb{R}} |\widehat{Y_T}(\mu)|^2\,d\mu \longrightarrow 2$ *as* $T \to \infty$*, where* $\Theta(T) = \frac{1}{2}T\log(T/2\pi) + O(T)$.

Proof. Combine Lemma 5 with (HL): $\int_0^T |Z|^2\,dt = T\log(T/2\pi) + O(T) \sim 2\Theta(T)$. □

3 Mode factorization

Lemma 7 (Riemann–Siegel factorization of $\widehat{Y_T}$). *For* $T \geq 2$, $\mu \in \mathbb{R}$,

$$\widehat{Y_T}(\mu) = \int_0^T Z(t)\sqrt{\Theta'(t)}\,e^{-i\mu\Theta(t)}\,dt = \sum_{n=1}^{N(T)} \sum_{\sigma\in\{\pm1\}} \mathcal{J}_{n,\sigma}(T,\mu) + \mathcal{R}_T(\mu),$$

$$\mathcal{J}_{n,\sigma}(T,\mu) := n^{-1/2}\int_{t_n}^T \sqrt{\Theta'(t)}\,e^{i\Phi_{n,\sigma,\mu}(t)}\,dt, \qquad \Phi_{n,\sigma,\mu}(t) := \sigma(\vartheta(t) - t\log n) - \mu\Theta(t),$$

with $t_n := \max(2, 2\pi n^2)$.

Proof. Substitute $u = \Theta(t)$ in the Fourier integral: $du = \Theta'(t)dt$, $Y(u)du = Z(t)\Theta'(t)/\sqrt{\Theta'(t)}\,dt = Z(t)\sqrt{\Theta'(t)}\,dt$. Insert (RS) and expand $\cos = (e^{i\cdot} + e^{-i\cdot})/2$; the $\sigma = \pm1$ index tracks the two exponentials. Modes $n > N(t)$ do not contribute at time t, giving the cutoff $t \geq 2\pi n^2$. □

Lemma 8 (Phase derivative). *Let* $\omega_n(t) := (\vartheta'(t) - \log n)/\Theta'(t)$*. Then for* $t \geq t_n$*:*

(i) $\Phi'_{n,\sigma,\mu}(t) = \Theta'(t)\,(\sigma\omega_n(t) - \mu)$.

(ii) $\omega_n(t) \in [0,1)$.

(iii) For $|\mu| \geq 1+\epsilon$, $|\Phi'_{n,\sigma,\mu}(t)| \geq \epsilon\,\Theta'(t)$.

Proof. (i) Direct: $\Phi' = \sigma\vartheta' - \sigma\log n - \mu\Theta' = \Theta'(\sigma\omega_n - \mu)$. (ii) For $t \geq 2\pi n^2$, $\vartheta'(t) \geq \log n$, so $\omega_n \geq 0$; and $\Theta' - (\vartheta' - \log n) = c + \log n > 0$, so $\omega_n < 1$. (iii) $\sigma\omega_n \in [-1,1)$, so $|\sigma\omega_n - \mu| \geq |\mu| - 1 \geq \epsilon$. □

4 Band-limit bound

Lemma 9 (Single-mode off-band bound). *Fix* $\epsilon > 0$ *and* $|\mu| \geq 1+\epsilon$*. For each* $1 \leq n \leq N(T)$ *and* $\sigma \in \{\pm1\}$,

$$|\mathcal{J}_{n,\sigma}(T,\mu)| \leq \frac{C_1}{\epsilon\,n^{1/2}}\left(\frac{1}{\sqrt{\Theta'(T)}} + \frac{1}{\sqrt{\Theta'(t_n)}}\right) + \frac{C_2}{\epsilon^2\,n^{1/2}}.$$

Proof. Integration by parts with $e^{i\Phi} = (i\Phi')^{-1}(d/dt)e^{i\Phi}$:

$$\mathcal{J}_{n,\sigma} = n^{-1/2}\left[\frac{\sqrt{\Theta'}}{i\Phi'}e^{i\Phi}\right]_{t_n}^{T} - n^{-1/2}\int_{t_n}^{T}\frac{d}{dt}\left(\frac{\sqrt{\Theta'}}{i\Phi'}\right)e^{i\Phi}\,dt.$$

Boundary: $|\sqrt{\Theta'}/\Phi'| \le 1/(\epsilon\sqrt{\Theta'})$ by Lemma 8(iii).

Integrated part: $\frac{d}{dt}(\sqrt{\Theta'}/\Phi') = \Theta''/(2\sqrt{\Theta'}\Phi') - \sqrt{\Theta'}\Phi''/\Phi'^2$, with $\Phi'' = \sigma\vartheta'' - \mu\vartheta''$ (since $\Theta'' = \vartheta''$) so $|\Phi''| \le (1+|\mu|)|\vartheta''|$. Both pieces are bounded by $C|\vartheta''|/(\epsilon^2\Theta'^{3/2})$; integrating, $\int_2^\infty |\vartheta''|/\Theta'^{3/2}\,dt \le C\int_2^\infty dt/(t(\log t)^{3/2}) < \infty$. □

Lemma 10 (Remainder bound via Gabcke). *Let $\mathcal{R}_T(\mu) := \int_0^T R_{\mathrm{RS}}(t)\sqrt{\Theta'(t)}e^{-i\mu\Theta(t)}\,dt$. Uniformly in $\mu\in\mathbb{R}$,*

$$|\mathcal{R}_T(\mu)| \le \int_0^T |R_{\mathrm{RS}}(t)|\sqrt{\Theta'(t)}\,dt = O(T^{1/4}\sqrt{\log T}).$$

Proof. By (RS), $|R_{\mathrm{RS}}(t)| = O(t^{-3/4})$ for $t\ge 2$, and $\sqrt{\Theta'(t)} = O(\sqrt{\log t})$. Triangle inequality on the integral defining $\mathcal{R}_T$ gives

$$|\mathcal{R}_T(\mu)| \le C + C'\int_2^T t^{-3/4}\sqrt{\log t}\,dt = O(T^{1/4}\sqrt{\log T}),$$

uniformly in μ. □

Theorem 11 (Band-limit). *For every $\mu\in\mathbb{R}$ with $|\mu|>1$,*

$$\frac{|K_T(\mu)|^2}{\Theta(T)} = O_\mu\left(\frac{1}{T^{1/2}}\right) \longrightarrow 0, \qquad T\to\infty,$$

locally uniformly on $\{|\mu|>1\}$.

Proof. Fix $\epsilon>0$, $|\mu|\ge 1+\epsilon$. By Lemma 9,

$$\sum_{n=1}^{N(T)}\sum_{\sigma}|\mathcal{J}_{n,\sigma}| \le \frac{2C_1}{\epsilon}\sum_{n=1}^{N(T)}\frac{1}{n^{1/2}\sqrt{\Theta'(t_n)}} + \frac{2C_1}{\epsilon\sqrt{\Theta'(T)}}\sum_{n=1}^{N(T)}\frac{1}{n^{1/2}} + \frac{2C_2}{\epsilon^2}\sum_{n=1}^{N(T)}\frac{1}{n^{1/2}}.$$

Using $\Theta'(t_n)\asymp\log n$ for $n\ge 2$, $\Theta'(T)\asymp\log T$, $\sum_{n\le N} n^{-1/2} = O(\sqrt{N})$, $\sum_{n\le N} 1/(n^{1/2}\sqrt{\log n}) = O(\sqrt{N/\log N})$, the mode sum is $O_\epsilon(T^{1/4})$. Combined with the remainder bound $|\mathcal{R}_T(\mu)| = O(T^{1/4}\sqrt{\log T})$ of Lemma 10,

$$|\widehat{Y_T}(\mu)| = O_\epsilon(T^{1/4}\sqrt{\log T}).$$

Hence

$$\frac{|K_T(\mu)|^2}{\Theta(T)} = \frac{|\widehat{Y_T}(\mu)|^2}{(2\pi)^2\Theta(T)} = \frac{O_\epsilon(T^{1/2}\log T)}{T\log T/2} = O_\epsilon\left(\frac{1}{T^{1/2}}\right).$$

Uniformity on compact subsets of $\{|\mu|>1\}$ is immediate from the uniform lower bound on $\epsilon = |\mu|-1$. □

5 Weak-$*$ concentration of the spectral measure

Definition 12. $d\rho_T(\mu) := |\widehat{Y_T}(\mu)|^2/(2\pi\Theta(T))\, d\mu$.

Corollary 13. *The family $\{\rho_T\}_{T\geq 2}$ of nonnegative Borel measures on $\mathbb{R}$ has total mass $\rho_T(\mathbb{R}) \to 2$, and for every $\epsilon > 0$,*

$$\rho_T\big(\mathbb{R} \setminus [-1-\epsilon, 1+\epsilon]\big) \longrightarrow 0, \qquad T \to \infty.$$

Consequently, any weak-$*$ subsequential limit ρ is a nonnegative even Borel measure with $\operatorname{supp}\rho \subseteq [-1,1]$ *and* $\rho([-1,1]) = 2$.

Proof. Total mass $\rho_T(\mathbb{R}) \to 2$ by Corollary 6. Evenness of ρ_T from $|\widehat{Y_T}(-\mu)| = |\widehat{Y_T}(\mu)|$ (reality of Y).

Middle off-band window. Fix $\epsilon > 0$ and $M > 1+\epsilon$. By Theorem 11 with constants uniform on $[1+\epsilon, M]$,

$$\rho_T\big(\{1+\epsilon \leq |\mu| \leq M\}\big) \leq \int_{1+\epsilon\leq|\mu|\leq M} O_\epsilon(T^{-1/2})\, d\mu = O_\epsilon\left(\frac{M}{T^{1/2}}\right).$$

Tail $|\mu| > M$ via iterated integration by parts in u. Since Z is entire and Θ is a C^∞ diffeomorphism with $\Theta' \geq c - c_\star > 0$, the function Y is C^∞ on $\mathbb{R}$. Integration by parts k times in u on the integral defining $\widehat{Y_T}(\mu) = \int_0^{\Theta(T)} Y(u) e^{-i\mu u}\, du$ gives

$$|\widehat{Y_T}(\mu)| \leq \frac{C_k(T)}{|\mu|^k}, \qquad C_k(T) := \sum_{j=0}^{k-1} \big(|Y^{(j)}(0)| + |Y^{(j)}(\Theta(T))|\big) + \int_0^{\Theta(T)} |Y^{(k)}(u)|\, du.$$

Each $Y^{(j)}$ is bounded on $[0, \Theta(T)]$ by a polynomial in T: the convexity bound $|Z(t)| = O(t^{1/4+\delta})$ and its derivative analogues give $|Y^{(j)}(u)| = O(T^{1/4+\delta})$ for $u \leq \Theta(T)$, hence $C_k(T) = O(T^{5/4+\delta})$ for every fixed $k \geq 1$.

Therefore, with $k \geq 2$ fixed,

$$\int_{|\mu|>M} |\widehat{Y_T}(\mu)|^2\, d\mu \leq \frac{2\, C_k(T)^2}{(2k-1)\, M^{2k-1}}, \qquad \rho_T\big(\{|\mu| > M\}\big) \leq \frac{C_k(T)^2}{\pi\,(2k-1)\,\Theta(T)\, M^{2k-1}}.$$

Cutoff choice. Set $M = M(T) := T^{1/4}$ and $k := 4$. Middle off-band: $O_\epsilon(T^{1/4}/T^{1/2}) = O_\epsilon(T^{-1/4}) \to 0$. Tail: $C_4(T)^2 = O(T^{5/2+2\delta})$, $M^{2k-1} = T^{7/4}$, $\Theta(T)\, M^{2k-1} \asymp T\log T \cdot T^{7/4} = T^{11/4}\log T$, giving $\rho_T(\{|\mu| > T^{1/4}\}) = O(T^{5/2+2\delta}/T^{11/4}\log T) = O(T^{-1/4+2\delta}/\log T) \to 0$. With $M = T^\alpha$ and $\alpha \in (0, 1/2)$, the middle rate is $T^{\alpha-1/2}$ and the tail rate is $T^{3/2-\alpha(2k-1)+2\delta}/\log T$; balancing gives $\alpha = 1/k$, so $(\alpha, k) = (1/4, 4)$ is the joint optimum at this k, with common rate $T^{-1/4+2\delta}$.

Hence $\rho_T(\{|\mu| > 1+\epsilon\}) \to 0$ for every $\epsilon > 0$.

Weak-$*$ limit. Banach–Alaoglu extracts a weak-$*$ convergent subsequence $\rho_{T_k} \rightharpoonup \rho$. For $\varphi \in C_c(\mathbb{R} \setminus [-1,1])$ supported in $\{|\mu| > 1+\epsilon\}$ for some $\epsilon > 0$, $\int \varphi\, d\rho = \lim \int \varphi\, d\rho_{T_k} = 0$ (since total off-band mass $\to 0$ dominates any bounded continuous test function supported there). Hence $\operatorname{supp}\rho \subseteq [-1,1]$. Total mass in the limit: $\rho([-1,1]) = \lim \rho_{T_k}([-1,1]) = \lim(\rho_{T_k}(\mathbb{R}) - \rho_{T_k}(\{|\mu| > 1\})) = 2 - 0 = 2$. Evenness passes from ρ_T to ρ. $\square$

References

[1] H.M. Edwards, *Riemann's Zeta Function*, Academic Press, 1974 (Riemann–Siegel main formula).

[2] W. Gabcke, *Neue Herleitung und explizite Restabschätzung der Riemann-Siegel-Formel*, Doctoral thesis, Georg-August-Universität Göttingen, 1979 (reissued 2015); establishes $|R_{\mathrm{RS}}(t)| = O(t^{-3/4})$ with explicit constants for the second-order Riemann–Siegel formula.

[3] E.C. Titchmarsh, *The Theory of the Riemann Zeta-Function*, 2nd ed., Oxford, 1986 (Hardy–Littlewood second moment, ϑ-asymptotics).